Deployment Report

1. Frontend Hosting (Vercel)

Deployed frontend application - https://vercel.com

Platform Configuration

The frontend is a Single Page Application (SPA) built with React and Vite, automatically deployed via GitHub integration.

Setting	Value
Framework Preset	Vite
Build Command	npm run build
Install Command	npm install
Output Directory	dist
Root Directory	frontend

Environment Variables

Variable Key	Production Value
VITE_API_URL	https://energymate-api-production.up.railway.app/api

Routing Rules (vercel.json)

To prevent 404 errors on direct page reloads, a rewrite rule is configured in the frontend root to direct all traffic to the React router:

```
{
	"rewrites": [
		{ "source": "/(.*)", "destination": "/index.html" }
	]
}
```

2. Backend Hosting (Railway)

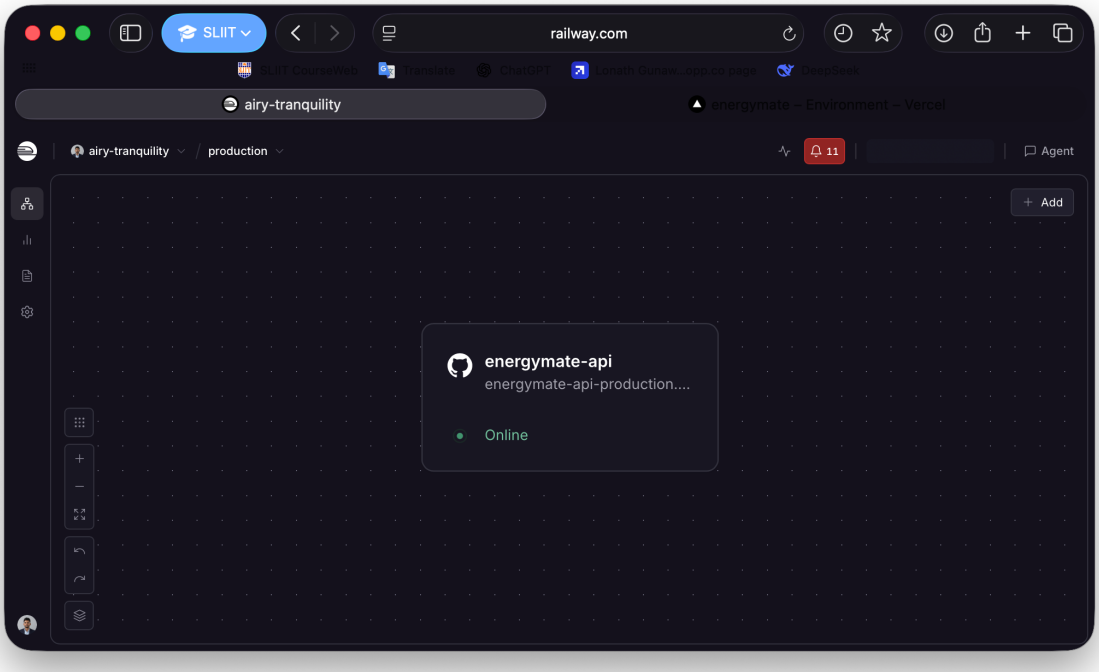
Deployed backend API - https://railway.com

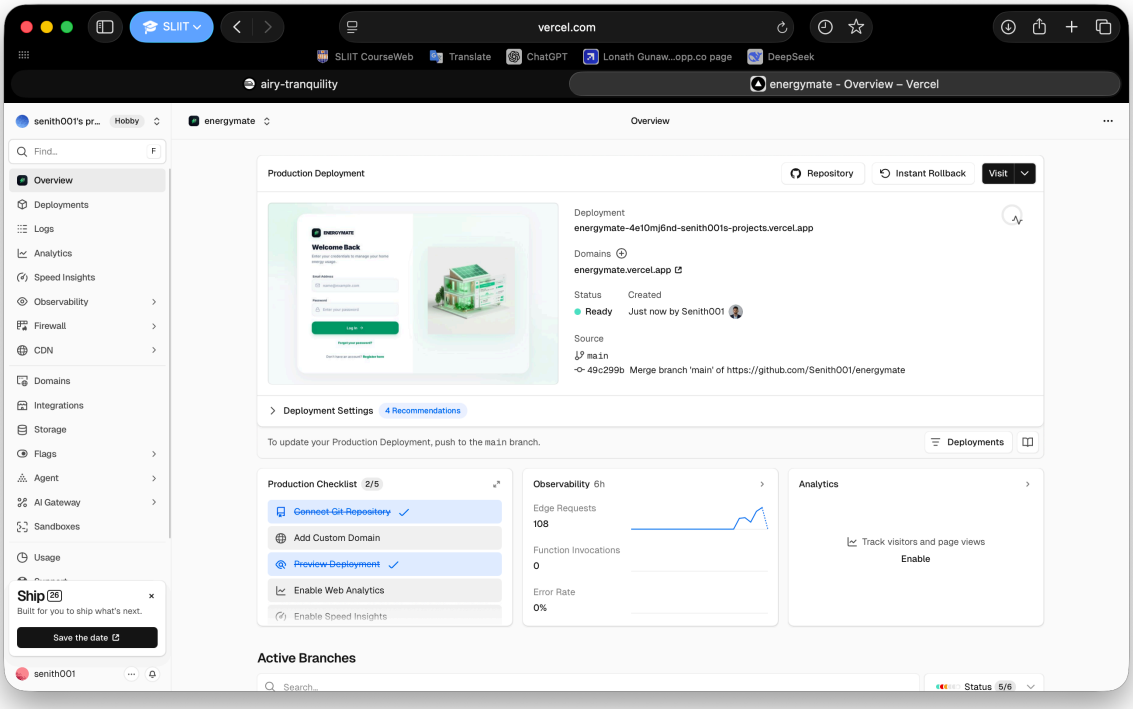
Platform Configuration

The backend is a Node.js REST API using Express and MongoDB. It is deployed as a continuousdelivery service connected to the repository.

Setting	Value
Environment	Node.js
Start Command	npm start
Public Networking	Enabled (Port 5001)
Root Directory	backend

Screenshots of successful deployment





Testing Report

1. Running Backend Tests

Open terminal and navigate to the backend directory:

```
cd backend
```

Run the commands below based on what you need:

```
npm test
```

Run tests in Watch Mode:

```
npm run test:watch
```

Generate a Code Coverage Report:

```
npm run test -- --coverage
```

2. Running Frontend Tests

Open a new terminal and navigate to the frontend directory:

```
cd frontend
```

Run all tests once:

```
npm test
```

Run the Interactive UI Mode:

```
npm run test:ui
```